



ANDERSON JUNIOR COLLEGE
2018 JC 2 PRELIMINARY EXAMINATIONS

CHEMISTRY

Paper 3 Free Response

9729/03

14 September 2018

2 hours

Candidates answer on separate paper.

Additional Materials: Answer Paper
 Graph Paper
 Data Booklet

READ THESE INSTRUCTIONS FIRST

Write your name, PDG and register number on all the work you hand in.

Write in dark blue or black pen.

You may use a pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer **one** question.

The use of an approved scientific calculator is expected, where appropriate.
A Data Booklet is provided.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

This document consists of **12** printed pages.

Section A

Answer **all** the questions in this section.

- 1 Aqua regia is a mixture of nitric and hydrochloric acid and can be used to dissolve noble metals such as gold. Although gold is typically an inert metal and it does not dissolve in either of the acids alone, it will dissolve in aqua regia because of the unique combined action of the two acids.

The standard reduction potential of Au^{3+} to Au is given below.

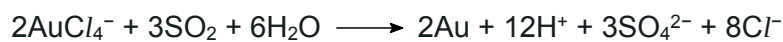


- (a) (i) With the use of relevant data from the *Data Booklet*, explain why hydrochloric acid alone cannot dissolve gold. [1]

In aqua regia, nitric acid is reduced to nitrogen dioxide when it converts gold to its ionic form, Au^{3+} and hydrochloric acid provides a source of chloride which react with the gold cations to form stable tetrachloroaurate(III) anions, AuCl_4^- .

- (ii) Write the balanced equation for the reaction between nitric acid and gold occurring in aqua regia. [1]
- (iii) Explain, with the use of suitable equations, how the presence of hydrochloric acid encourages the reaction in (a)(ii) to take place readily. [2]

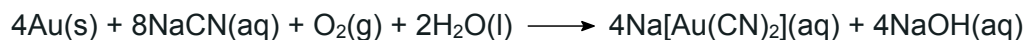
- (b) If the aqua regia solution only contains gold, it may be reduced by sulfur dioxide to obtain elemental gold.



Gold ore from a particular mining site is said to contain about 4.0 g of gold in one tonne of ore. In three months, the site processed 2.18×10^6 tonnes of ore. Determine the minimum volume of sulfur dioxide gas, measured at room temperature and pressure, required to extract elemental gold in three months.

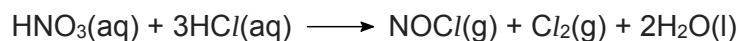
[1 tonne = 1000 kg] [3]

- (c) Another method of extracting gold from low-grade ore involves converting the gold to a water-soluble dicyanoaurate(I) anionic complex, $[\text{Au}(\text{CN})_2]^-$.

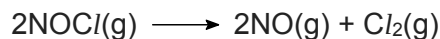


- (i) State the shape of the dicyanoaurate(I) anionic complex. [1]
- (ii) Explain why the gold complex ion, AuCl_4^- in (a) is coloured while the dicyanoaurate(I) complex ion, $[\text{Au}(\text{CN})_2]^-$, is colourless. [4]

- (d) Nitrosyl chloride, NOCl , is a yellow gas that is most commonly encountered as a decomposition product of aqua regia.



Nitrosyl chloride can further decompose into nitric oxide and chlorine.



The decomposition of gaseous NOCl was studied at 125°C and Table 1.1 shows the concentration of NOCl at various times.

Table 1.1

time / s	concentration of NOCl / molecule cm^{-3}
0	1.5×10^{23}
1.1×10^3	1.2×10^{23}
2.9×10^3	0.9×10^{23}
5.8×10^3	0.6×10^{23}
13.8×10^3	0.3×10^{23}

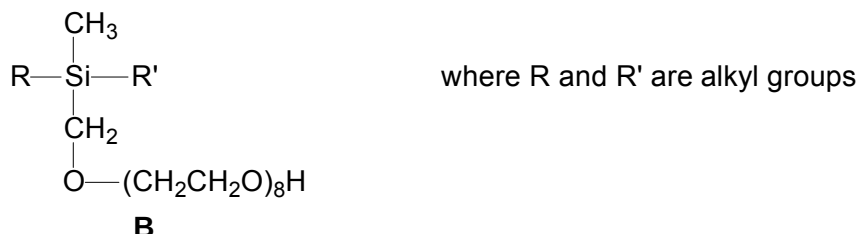
- (i) Suggest why the aqua regia solution is best used to dissolve noble metals immediately after it is made. [1]
- (ii) Use the data given in Table 1.1 to plot a graph of concentration of NOCl against time on the graph paper provided. [2]
- (iii) Hence, use your graph to show that the decomposition of NOCl is a second order reaction. [2]
- (iv) Calculate the rate constant of the reaction in $\text{mol}^{-1} \text{dm}^3 \text{s}^{-1}$. [2]
- (v) It has been found that addition of nitrogen dioxide, NO_2 leads to a faster decomposition of nitrosyl chloride at the same temperature, during which the nitrogen dioxide concentration remains unchanged.

With an appropriate sketch of Boltzmann distribution, explain why the addition of nitrogen dioxide speeds up the rate of decomposition. [2]

[Total: 21]

- 2 (a) Oil spills are ecological disasters and remediating them can run into billions of dollars. One of the methods adopted to combat oil spills was the use of dispersants, which break up the oil into tiny droplets and help to spread out the droplets throughout the water.

An example of a chemical dispersant **B**, which contains both carbon and silicon, has the following structure.



- (i) Draw a diagram to illustrate hydrogen bonding between a molecule of **B** and a water molecule. [1]
- (ii) By reference to the type of intermolecular forces, explain why this molecule of **B** is able to function as an oil spill dispersant. [2]
- (b) Both carbon and silicon are among the elements of Group 14. It was observed that those towards the top, carbon to germanium, have very different properties from those at the bottom, tin and lead.

Table 2.1 shows the marked change in melting points after germanium.

Table 2.1

element	C	Si	Ge	Sn	Pb
mp / °C	>3550	1410	937	232	327

- (i) What type of structure is present in solid carbon, silicon and germanium? [1]
- (ii) Suggest why the melting points of these elements decrease from carbon to germanium. [1]
- Carbon and silicon each form a tetrachloride. CCl_4 has no reaction with water; SiCl_4 reacts readily with water.
- (iii) Suggest an explanation for this observation. [1]
- (iv) Hence, suggest a balanced equation for the reaction between GeCl_4 and water and state the colour observed when a few drops of Universal Indicator solution was added to the resulting mixture. [2]

- (c) Friedel–craft alkylation, using AlCl_3 as a catalyst, is usually carried out in anhydrous conditions. In the presence of AlCl_3 and under suitable conditions, benzene reacts with CCl_4 to form dichlorodiphenylmethane, $(\text{C}_6\text{H}_5)_2\text{CCl}_2$.

- (i) Suggest why the reaction has to be carried out under anhydrous condition. [1]
- (ii) Name and describe the mechanism of the reaction, showing any intermediates. [4]
- (iii) Suggest a reason why further alkylation of the product in (c)(ii) does not take place. [1]

Fig. 2.1 shows two examples of Friedel–craft alkylation which can be done in non–anhydrous conditions.

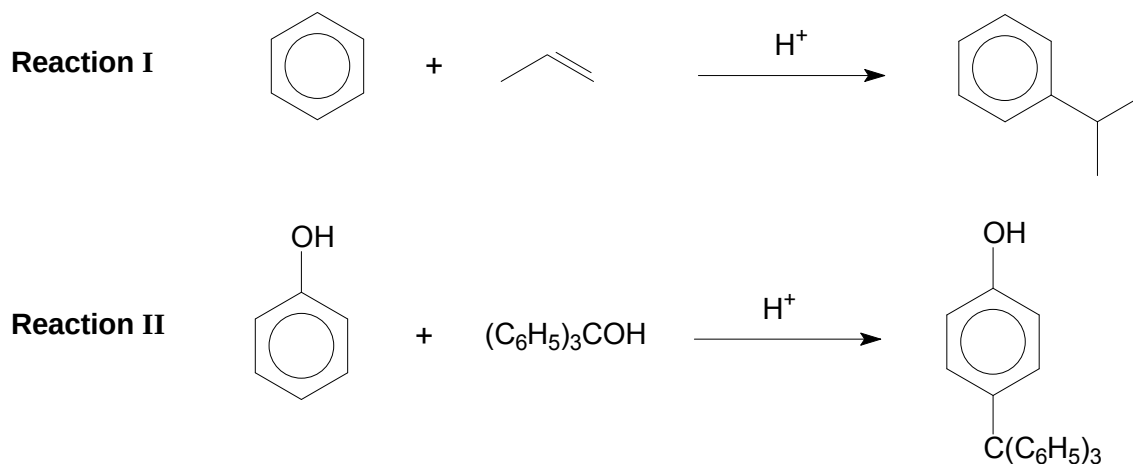
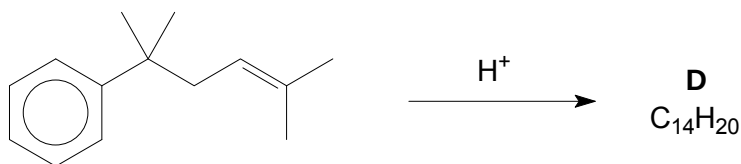
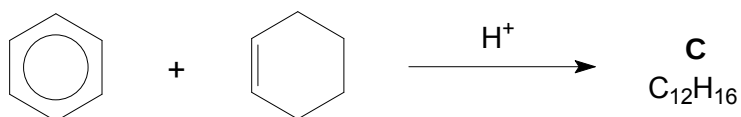


Fig. 2.1

- (iv) Using the information given about **Reaction I**, predict the structure of the product of each of the following reactions.

Both compounds **C** and **D** do not react with aqueous bromine.



[2]

- (v) Hence, suggest the reagents and conditions for a reaction that could be used to distinguish between **C** and **D**.
Draw the structure of the product(s) of any reactions that occur. [2]

- (vi) **Reaction II cannot** take place if benzene is used instead of phenol.

With reference to the structure of the reactant and the intermediate formed, suggest why this is so. [2]

[Total: 20]

- 3 (a) The following scheme shows the synthesis of aspirin using the Kolbe process. Salicylic acid is formed as an intermediate. Study Fig. 3.1 carefully and answer the questions that follow.

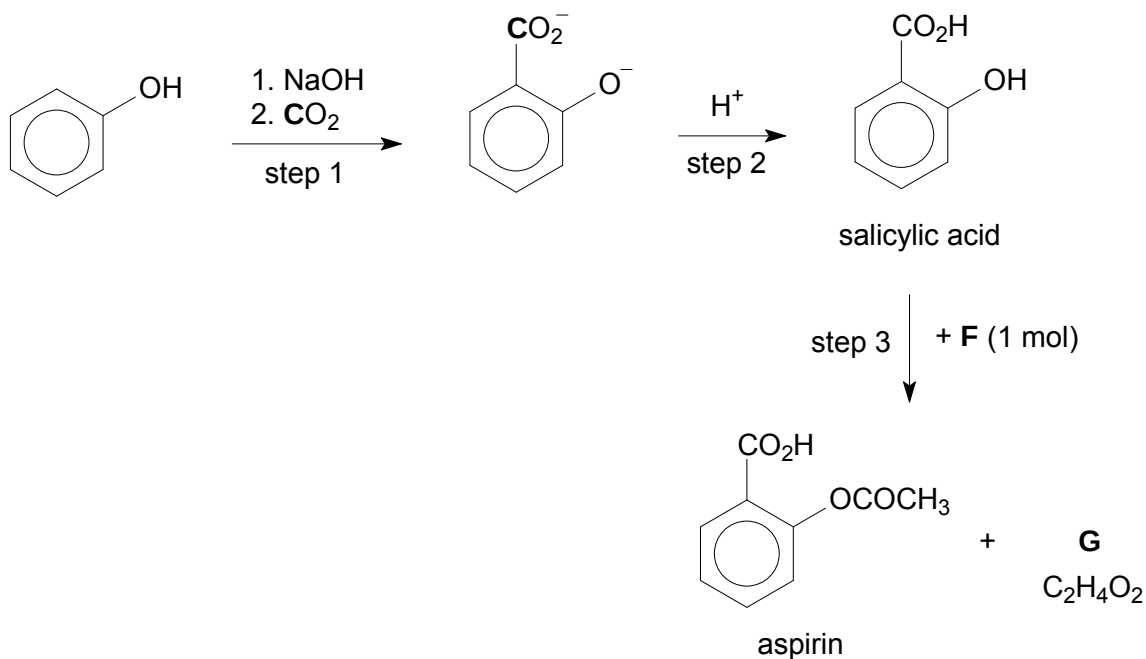
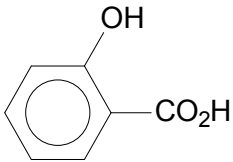
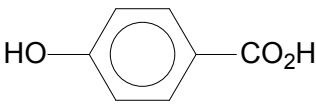


Fig. 3.1

- (i) State the change in type of hybridisation with respect to the carbon atom (in **bold**), from CO_2 to the product of step 1. [1]
- (ii) Suggest the type of reaction that occur in step 2. [1]
- (iii) **G** forms a soluble salt with NaOH(aq) .
Suggest the identity of the product **G** formed in step 3. [1]
- (iv) Hence, suggest the molecular formula of the reagent **F** in step 3. [1]

- (b) Table 3.1 shows the boiling point and pK_{a1} values of salicylic acid and its structural isomer, 4-hydroxybenzoic acid.

Table 3.1

compound	melting point / °C	pK_{a1}
 salicylic acid	160	3.0
 4-hydroxybenzoic acid	214	4.5

- (i) Explain why the melting point of salicylic acid is lower than that of 4-hydroxybenzoic acid. [2]
- (ii) Suggest a reason why the pK_{a1} value of salicylic acid is less than that of 4-hydroxybenzoic acid. [2]
- (c) Depending on the conditions of the reaction, 4-ethylphenol can react with chlorine in two different ways, giving the two isomers **P** and **Q** as shown in Fig. 3.2.

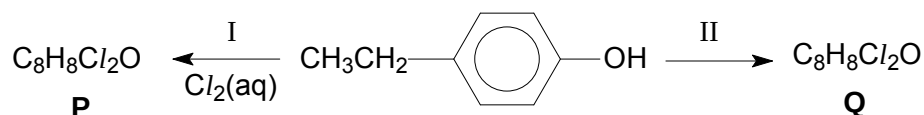


Fig. 3.2

- (i) Suggest a structural formula for compound **P**. [1]
- (ii) Explain whether you would expect **P** to be more or less acidic than 4-ethylphenol. [2]

Compound **Q** reacts with $\text{NaOH}(\text{aq})$, followed by acidification to give **R**, $\text{C}_8\text{H}_{10}\text{O}_3$, which immediately loses water to form **S**, $\text{C}_8\text{H}_8\text{O}_2$. Compound **S** is insoluble in water but it dissolves in $\text{NaOH}(\text{aq})$. **S** reacts with 2,4-dinitrophenylhydrazine and with alkaline aqueous iodine, but not with Tollens' reagent.

- (iii) Suggest structures for **Q**, **R** and **S**. Explain the reactions described. [5]
- (iv) Write a balanced equation for each of the reaction between **S** and
- 2,4-dinitrophenylhydrazine
 - alkaline aqueous iodine
- [2]
- (v) Suggest suitable conditions for reaction II. [1]

[Total: 19]

Section B

Answer **one** question from this section.

- 4 (a) The hydrogen fuel cell gives green energy but there are problems associated with hydrogen sources and storage, limiting its use in both transportation and stationary applications. While fuel cells which use liquid alcohols pose no storage problem, they emit carbon dioxide and thus countering the move to reduce carbon footprint.

As a result, attention has turned to other alternative fuels such as ammonia and hydrazine, which on combustion gives nitrogen and water. Fig 4.1 shows a prototype of an ammonia alkaline fuel cell using aqueous potassium hydroxide membrane.

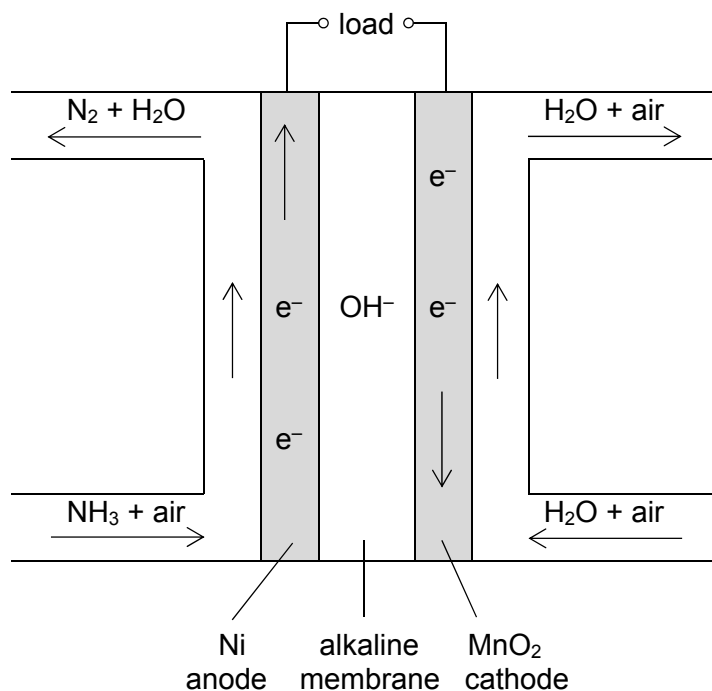
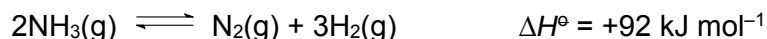


Fig. 4.1

At the anode, ammonia is oxidised to nitrogen while oxygen is reduced at the cathode.

- (i) Write the equations for the reactions occurring at the anode and cathode. [2]
- (ii) Hence, write the equation for the electro-combustion of one mole of ammonia. [1]
- (iii) The value of ΔG° at 298 K for the equation you have written in (a)(ii) is -339 kJ mol^{-1} . Calculate the value for E°_{cell} for the ammonia alkaline fuel cell. [2]
- (iv) Using suitable data from the *Data Booklet*, suggest a value for the E° of the N_2/NH_3 electrode reaction. [1]
- (v) Earlier prototypes of the ammonia fuel cell using the PEM (proton exchange membrane) was not viable. Suggest a reason for the limitation. [1]

- (b) Ammonia, in aqueous or liquid form, is promising as an energy medium for hydrogen storage and generation. Under suitable conditions, ammonia decomposes to nitrogen and hydrogen gases.



- (i) Write an expression for the equilibrium constant, K_c , for this reaction, and state its units. [2]

A mixture of 0.5 mol of N_2 and 2.0 mol of NH_3 was introduced into a sealed 2.00 dm³ vessel and heated to 670 K. At equilibrium, it was found that the amount of N_2 present was 1.2 mol.

- (ii) Calculate the equilibrium concentration of the reactant and the two products. [2]
 (iii) Use the data to calculate the value of K_c . [1]

For any reaction, ΔG° and the equilibrium constant, K_c are related according to the equation

$$\Delta G^\circ = -RT \ln K_c$$

- (iv) Use this equation and your answer to (b)(iii) to calculate the ΔG° of the reaction at 670 K. [1]
 (v) Hence, calculate the ΔS° of the reaction at 670 K and comment on its sign with reference to the reaction. [2]
 (vi) State and explain the effect on the position of equilibrium and K_c when the volume of the vessel was expanded to double the original volume at constant temperature. [2]
- (c) Ammonia reacts as a nucleophile with alkyl halides to give primary amines.

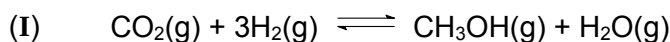
Describe the mechanism for the reaction between ammonia and methyl iodide. Show relevant lone pairs and dipoles, and use curly arrows to indicate the movement of electron pairs. [3]

[Total: 20]

- 5 (a) Chemical recycling of carbon dioxide to produce renewable materials and fuels, primarily methanol, offers a powerful alternative to tackle both issues of global climate change and fossil fuel depletion. Methanol is a colourless and flammable liquid which boils at 65 °C.

In Iceland, renewable methanol is formed by reacting carbon dioxide released naturally in the atmosphere from the geothermally active areas with hydrogen over a heterogeneous catalyst.

Using copper-based catalysts, methanol can be synthesised from CO₂ as shown in (I) below.



Some relevant enthalpy change values are given in Table 5.1.

Table 5.1

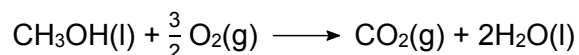
enthalpy change of combustion of H ₂ (g)	−286 kJ mol ^{−1}
enthalpy change of combustion of CH ₃ OH(g)	−768 kJ mol ^{−1}
enthalpy change of vapourisation of H ₂ O(l)	+40.8 kJ mol ^{−1}

Use the data in Table 5.1 to construct an energy cycle relating the reactants and products of the forward reaction in (I).

Hence, calculate the standard enthalpy change of the forward reaction in (I), $\Delta H_f^\ominus$. [3]

- (b) Research has shown that the direct methanol fuel cell (DMFC) is an appropriate alternative to rechargeable battery.

The overall cell reaction of a DMFC can be expressed as follow.



In a DMFC, methanol and water reacts at the anode to produce carbon dioxide and protons. The protons then move through an acidic conducting polymer electrolyte towards the cathode where they react with oxygen to produce water.

The electrons are transported through an external circuit from anode to cathode, providing power to connected devices like mobile phones and lap-top computers.

- (i) Write the two half equations at the electrodes which results in the occurrence of the cell reaction in a typical DMFC. [2]
- (ii) Explain why an alkaline electrolyte cannot be used in a DMFC. [1]
- (iii) A DMFC is capable of producing a voltage of 1.18 V at standard conditions.

Hence, calculate the standard Gibbs free energy change, $\Delta G^\ominus$, for the given overall cell reaction. [1]

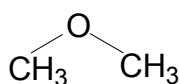
- (iv) Some relevant standard enthalpy change of formation values are given in Table 5.2.

Table 5.2

compound	$\Delta H_f^\ominus / \text{kJ mol}^{-1}$
$\text{CH}_3\text{OH}(\text{l})$	-238
$\text{CO}_2(\text{g})$	-394
$\text{H}_2\text{O}(\text{l})$	-286

Use the data given to calculate the enthalpy change of the overall cell reaction of DMFC at standard conditions. [1]

- (v) Hence, determine the standard entropy change, $\Delta S^\ominus$, for the overall cell reaction. [1]
- (vi) By writing appropriate equation(s), state and explain how the entropy change for the overall cell reaction will differ from your answer in (b)(v) if the fuel cell is operating at 80 °C. [2]
- (c) Beside its application in DMFC, methanol is also an important raw material in the production of dimethyl ether, DME.



DME

DME shows great potential as a clean high-efficiency compression ignition fuel. Its production from methanol involves the reaction between methanol and chloromethane via a nucleophilic substitution mechanism.

Step 1: Generation of a nucleophile through an acid-base reaction between methanol and a strong base, such as sodium hydroxide.

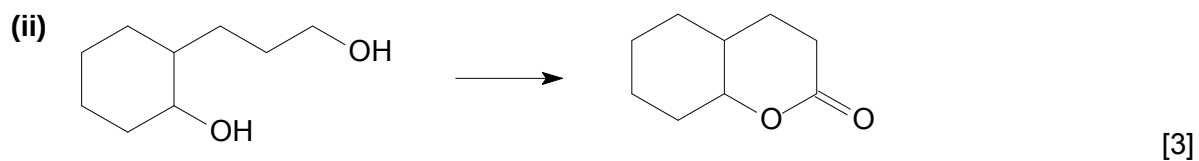
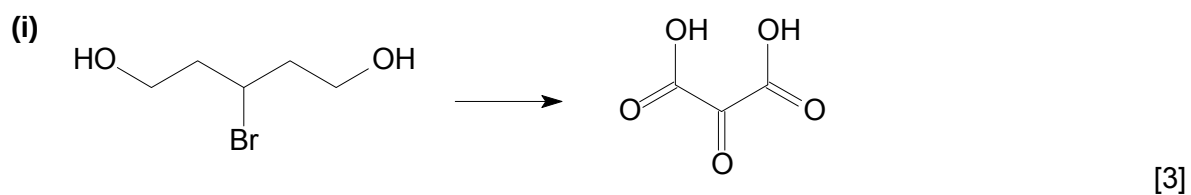
Step 2: Reaction of the nucleophile with chloromethane to form DME.

- (i) Write an ionic equation to show how the nucleophile is formed in **Step 1**. [1]
- (ii) Write equation(s) to show how the nucleophile generated in (c)(i) reacts with chloromethane in **Step 2**.

In your answer show any relevant charges, dipoles or lone pairs of electrons you consider to be important in this step. [2]

- (d) In **no more than 3 steps**, suggest how each of the following hydroxyl compounds can be transformed into the respective products.

You are required to state the reagents and conditions for each step and give the structures of the intermediate compounds.



[Total: 20]